Proposed Solution — RxAide: An AI-Powered Medication Reminder & Management Assistant

CSE299 – Section 4 | Group 2 | Faculty Advisor: Dr. Mohammad Shifat-E-Rabbi (MSRb)

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I. Introduction

Medication non-adherence is a critical healthcare challenge affecting 50% of patients on long-term therapies worldwide, contributing to approximately 125,000 preventable deaths annually in the United States and generating an estimated $100–289 billion in avoidable healthcare costs. Elderly patients and those managing polypharmacy regimens are at the highest risk due to the cognitive burden of complex medication schedules [1].

Our literature review [1] of seven existing medication management applications identified three critical gaps in the current landscape:

1. **The Integration Gap:** No existing application provides a unified, end-to-end pipeline from prescription image capture through data extraction, automatic schedule generation, reminders, and adherence tracking. Users are forced to either enter all data manually or combine multiple separate tools.
2. **The Accessibility Gap:** Manual data entry—required by every major reminder application (Medisafe, MyTherapy, Smart Meds Reminder)—disproportionately burdens elderly patients, those with limited technical literacy, and individuals managing complex multi-drug regimens.
3. **The Intelligence Gap:** While basic timer-based reminders are abundant, context-aware conversational assistance remains virtually absent. Only DoseMed integrates an AI assistant, and its scope is limited to medication queries rather than full-spectrum schedule management through natural language.

This report presents RxAide, an Android application designed to address all three identified gaps through an integrated, AI-powered medication management experience.

II. Proposed Solution

A. Solution Overview

RxAide is an intelligent medication management application that combines AI-powered prescription scanning, automated schedule generation, reliable reminders, dose adherence tracking, and a conversational AI assistant within a single, accessible platform. The application leverages the Google Gemini API [2] for both multimodal prescription OCR and conversational AI, eliminating the need for tedious manual data entry and providing an intuitive, end-to-end medication management workflow.

B. Core Capabilities

RxAide integrates five core capabilities—a combination that no existing application offers:

AI-Powered Prescription Scanning. Users photograph their prescription (handwritten or printed), and the Gemini API extracts medication names, dosages, frequencies, durations, and instructions in a single multimodal inference call. Google Search grounding is enabled to cross-reference and verify extracted medication names against pharmaceutical databases, reducing extraction errors. The system supports Bangla/English bilingual prescriptions and Bangladeshi dosage notation (e.g., "1+1+0" indicating morning, noon, and night doses).

Automated Schedule Generation. Upon user confirmation of extracted prescription data, RxAide automatically generates a personalized medication schedule with assigned time slots based on the prescribed frequency and user preferences. This capability is entirely absent from every application reviewed in our literature survey—all existing apps require manual schedule configuration, even those supporting OCR.

Reliable Medication Reminders. Notifications are delivered at each scheduled dose time using Android WorkManager, which ensures reliability even under battery optimization constraints and system restarts. Each notification includes action buttons allowing users to mark a dose as "Taken" or "Missed" directly from the notification without opening the application.

Dose Tracking and Adherence Analytics. The application maintains a comprehensive dose history, recording every dose event with timestamp and status (taken, missed, skipped). A visual dashboard provides adherence statistics over daily, weekly, and monthly intervals, giving users and caregivers clear insight into medication habits.

Conversational AI Assistant. A Gemini-powered chat interface enables natural language interaction for medication management. The system uses the user's medication schedule and recent chat history (last 20 messages) as context, enabling coherent multi-turn conversations. Users can ask questions about their medications, request schedule changes, and receive safe, general health guidance. All AI responses include explicit safety guardrails—the system provides only generic health information and recommends consulting healthcare professionals for specific medical advice.

III. System Architecture

RxAide follows the MVVM (Model-View-ViewModel) architecture pattern, which provides clean separation between the user interface, business logic, and data layers. The system is organized in three layers:

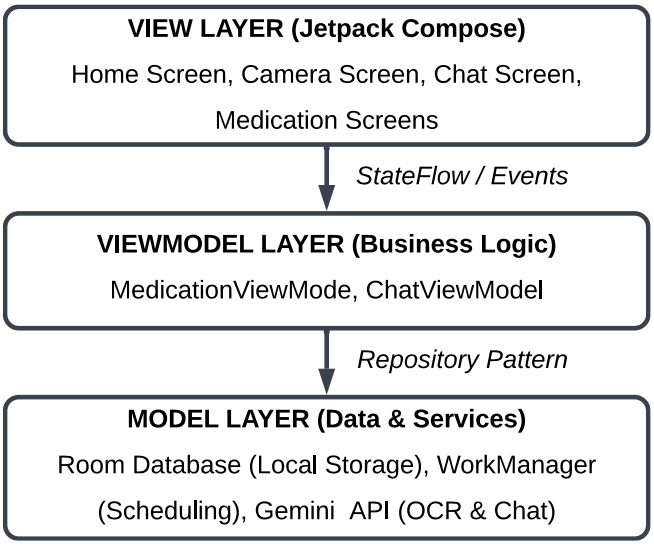

Fig. 1. RxAide three-layer MVVM system architecture.

- **View Layer:** Built with Jetpack Compose, providing declarative, responsive UI across six primary screens: Home, Camera (CameraX-powered), Chat, Add Medication, Medication List, and Medication Detail.
- **ViewModel Layer:** Contains business logic and manages data flow between the View and Model layers using Kotlin StateFlow and Coroutines for reactive, lifecycle-aware UI updates.
- **Model Layer**: Comprises Room for local data persistence (medications, schedules, dose history, chat messages), WorkManager for reliable background notification scheduling, and the Gemini API [2] for OCR processing and conversational AI.

IV. User Workflow

The primary user workflow follows five stages, forming the end-to-end pipeline that directly addresses the Integration Gap:

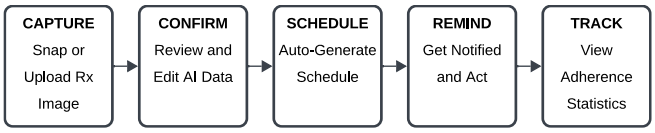

Fig. 2. RxAide five-stage user workflow pipeline.

1. **Capture Prescription:** The user photographs their prescription using the CameraX-powered camera screen, or selects an existing image from their device gallery.
2. **Confirm Extracted Details:** The image is automatically sent to the Gemini API via the chat interface. The AI extracts all medication details, and the user reviews and edits them before confirmation.
3. **Generate Schedule:** Upon confirmation, RxAide automatically generates a personalized medication schedule with appropriate time slots.
4. **Receive Reminders:** WorkManager delivers notifications at each scheduled time. Users mark doses as taken or missed directly from the notification.
5. **Track Adherence:** The dashboard displays taken, missed, and skipped dose statistics, providing actionable insight into medication habits.

V. Key Innovation and Differentiation

A. The End-to-End Pipeline

The most significant innovation of RxAide is the seamless, end-to-end pipeline from prescription capture to adherence tracking. This pipeline eliminates the fragmented workflow that currently forces users to combine separate OCR tools with separate reminder applications, or to manually enter all medication data.

B. Competitive Differentiation

Capability	Smart Meds	Prescription Reader AI	Medisafe	MyTherapy	DoseMed	RxAide
AI Prescription OCR	✗	✓	✗	✗	✗	✓
Auto Schedule Generation	✗	✗	✗	✗	✗	✓
Medication Reminders	✓	✗	✓	✓	✓	✓
Dose Tracking / Analytics	✓	✗	✓	✓	✓	✓
Conversational AI	✗	✗	✗	✗	✓	✓

The "Auto Schedule Generation" row is particularly notable—every cell is ✗ except RxAide. This represents the single most significant unmet need in the current medication management app market: no existing application automatically converts prescription data into a ready-to-use medication schedule.

VI. Expected Outcomes

Upon successful completion, RxAide will:

- Accurately extract medication details from both handwritten and printed prescription images using multimodal AI, supporting Bangla and English prescriptions.
- Automatically generate personalized medication schedules from confirmed prescription data, eliminating manual schedule configuration.
- Deliver reliable notifications for all scheduled doses, with direct take/miss actions from the notification.
- Track adherence with visual dashboards providing daily, weekly, and monthly statistics on medication compliance.
- Provide conversational assistance through a context-aware AI chat interface for medication queries and schedule management.

By eliminating the manual data entry barrier and providing an intelligent, unified workflow, RxAide aims to make medication management accessible to users of all ages and technical abilities—particularly elderly patients and those managing complex multi-drug regimens who are most at risk of non-adherence.

References

[1] RxAide Project Group 2, "Literature Review Presentation: AI-Powered Medication Reminder and Management Applications," CSE299 – Section 4, Spring 2026.

[2] Google AI, "Gemini API Documentation," 2026. Available: https://ai.google.dev/gemini-api/docs